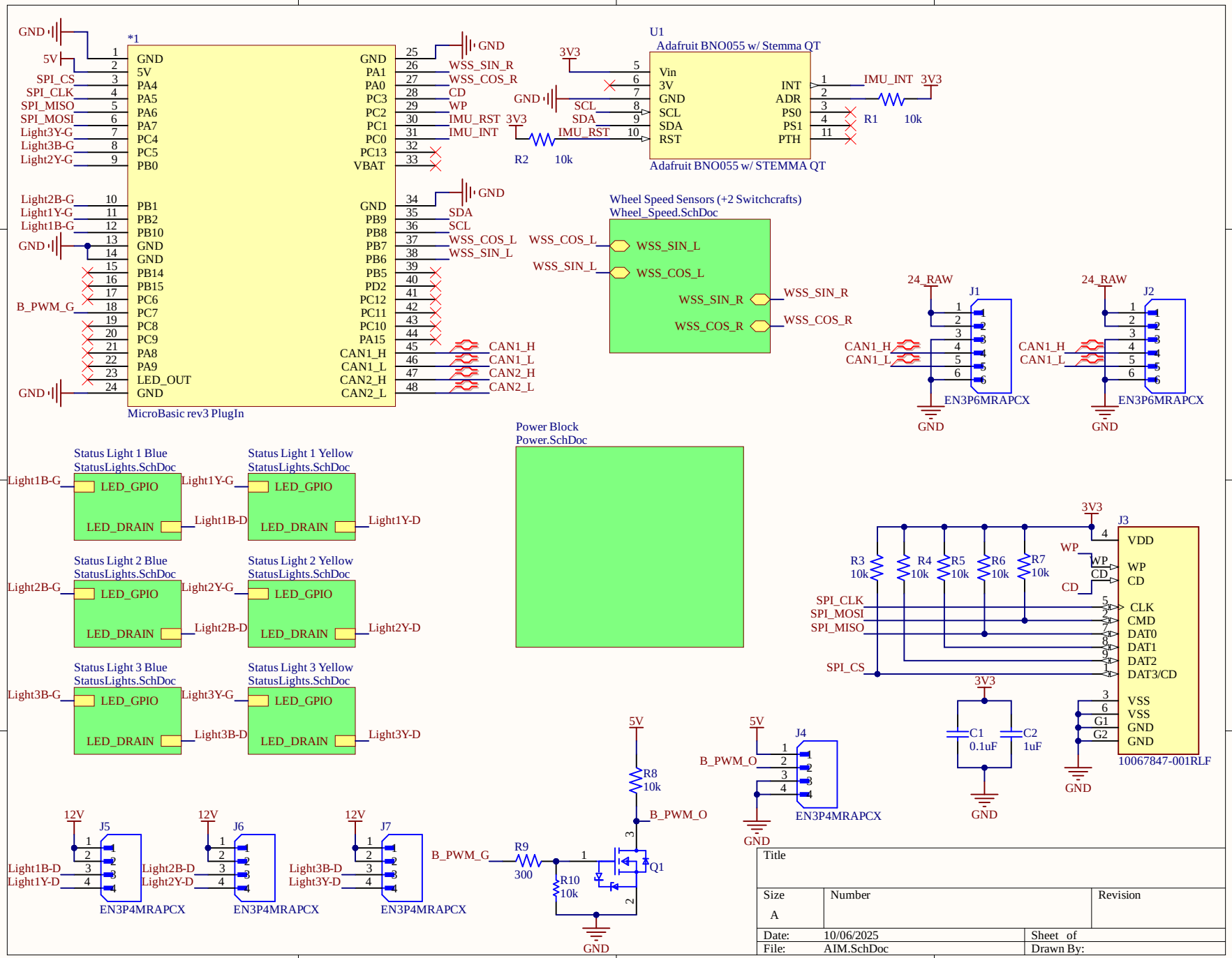

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Date: 10/06/2025		Sheet of
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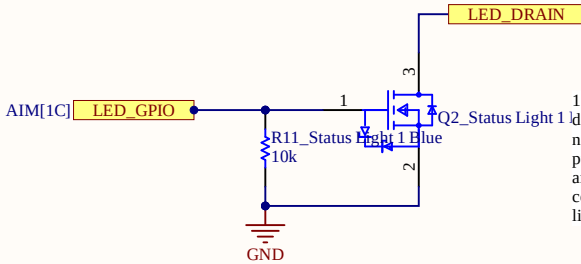
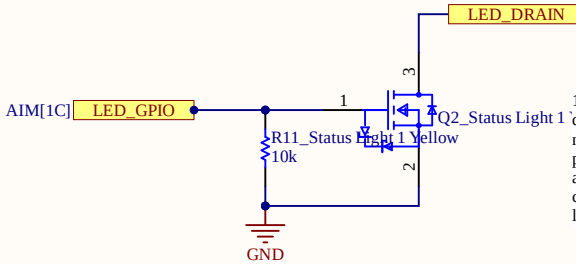


10k pulldown to have determinate state when MCU is not driving current. When MCU pulls line high, NMOS turns on and enables current flow through connectors, enabling offboard lights

Title		
Size A	Number	Revision
Date: 10/06/2025		Sheet of
File: StatusLights.SchDoc		Drawn By:



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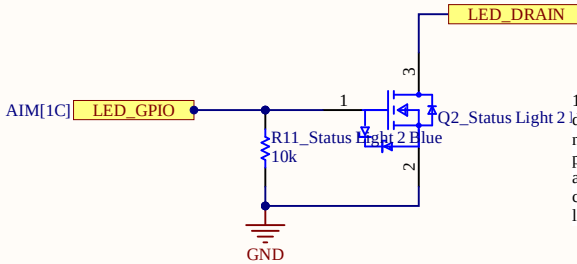
Title		
Size A	Number	Revision
Date: 10/06/2025		Sheet of
File: StatusLights.SchDoc		Drawn By:



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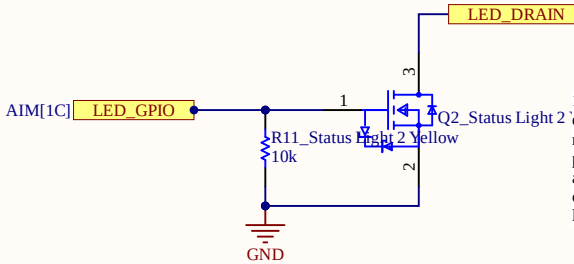
Date: 10/06/2025		Sheet of
File: StatusLights.SchDoc		Drawn By:



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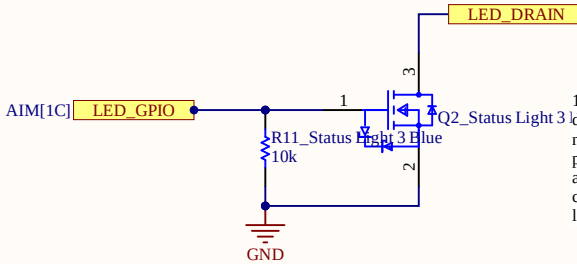
Size A	Number	Revision
Date: 10/06/2025		Sheet of
File: StatusLights.SchDoc		Drawn By:



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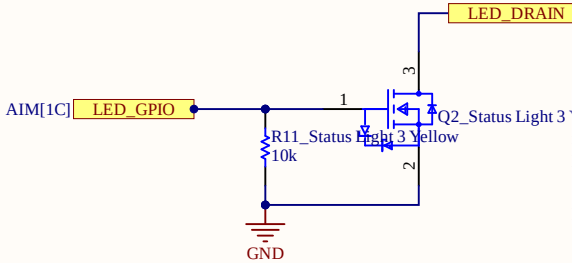
Date: 10/06/2025		Sheet of
File: StatusLights.SchDoc		Drawn By:



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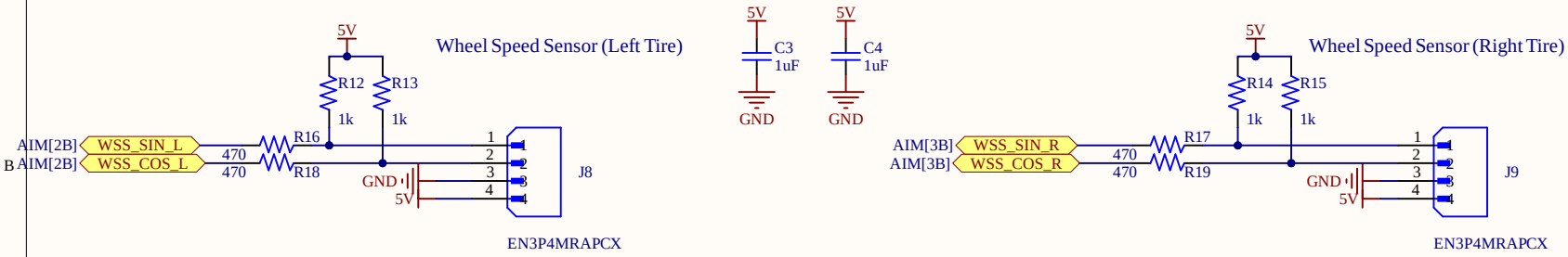
Title		
Size A	Number	Revision
Date: 10/06/2025		Sheet of
File: StatusLights.SchDoc		Drawn By:



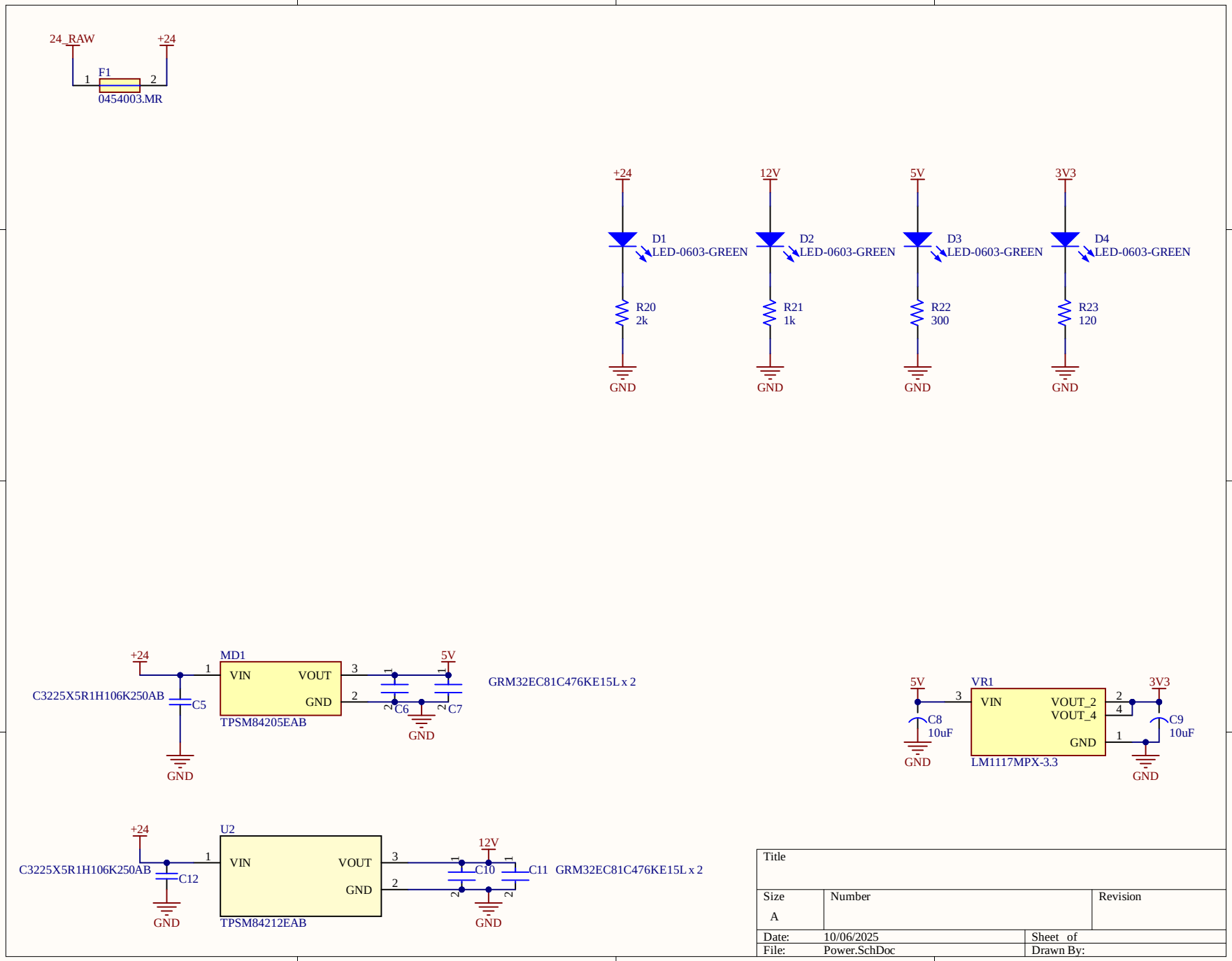
470 Ohm current limiting resistor so ~10mA sinks from sensor given by datasheet
1kOhm pull up resistor given by datasheet

Check for pull-up/pull-down

Check for pull-up/pull-down

Each Whel speed sensor outputs two square waves. Phase shift encodes direction, and frequency encodes speed. x freq Hz is x speed. x phase shift is x direction.

Title		
Size A	Number	Revision
Date: 10/06/2025		Sheet of
File: Wheel_Speed.SchDoc		Drawn By:


24_RAW
+24
F1
0454003.MR
+24
12V
5V
3V3
D1
LED-0603-GREEN
D2
LED-0603-GREEN
D3
LED-0603-GREEN
D4
LED-0603-GREEN
R20
2k
R21
1k
R22
300
R23
120
GND
MD1
VIN
VOUT
GND
TPSM84205EAB
C3225X5R1H106K250AB
C5
C6
C7
GRM32EC81C476KE15L x 2
VR1
VIN
VOUT_2
VOUT_4
GND
LM1117MPX-3.3
C8
10uF
C9
10uF
U2
VIN
VOUT
GND
TPSM84212EAB
C12
C10
C11
GRM32EC81C476KE15L x 2
Title
Size
A
Number
Revision
Date: 10/06/2025
Sheet of
File: Power.SchDoc
Drawn By: